

Chapter 13 Homework

Chapter 13 Homework – Tensors and Tensor Arithmetic

Part A – Python Skills Reinforcement

Problem 1 – Creating Tensors

Create the tensor:

```
T =  
[  
  [[1,2,3],[4,5,6]],  
  [[7,8,9],[10,11,12]]  
]
```

Tasks:

1. Create the tensor in NumPy.
2. Print T.shape.
3. Explain what each dimension means.
4. Determine:
 - Number of levels
 - Number of rows
 - Number of columns

Problem 2 – Tensor Indexing

Using T:

1. Print the first level.
2. Print the second level.
3. Print T[1,0,2].
4. Explain the indexing convention [level,row,column].

Problem 3 – Tensor Arithmetic

Create:

```
A =  
[  
  [[1,2],[3,4]],  
  [[5,6],[7,8]]  
]
```

```
B =  
[  
  [[10,20],[30,40]],  
  [[50,60],[70,80]]  
]
```

Compute:

1. A + B
2. A - B
3. A * B

4. A / B

Print shapes before every operation.

Problem 4 – Manual Tensor Arithmetic

Without using `A+B`, write nested loops that manually compute `C=A+B`.
Verify the result matches NumPy.

Problem 5 – Tensor Product

Define:

`a=[1,2,3]`

`b=[4,5]`

Compute:

`np.tensordot(a,b,axes=0)`

Tasks:

1. Print the result.
2. Print its shape.
3. Explain why the result is a matrix.

Part B – Practical Problem: Image Cube Analysis

Imagine a grayscale video represented as:

`Video[frame,row,column]`

Video =

```
[  
  [[10,11,12],[13,14,15],[16,17,18]],  
  [[20,21,22],[23,24,25],[26,27,28]],  
  [[30,31,32],[33,34,35],[36,37,38]],  
  [[40,41,42],[43,44,45],[46,47,48]]  
]
```

Tasks:

1. Compute the average brightness of each frame.
2. Compute the average brightness of the entire video.
3. Find the brightest pixel in the tensor.
4. Determine which frame contains the brightest pixel.
5. Multiply the entire tensor by 1.25 to simulate increasing brightness.
6. Explain why videos naturally require tensors rather than matrices.

Challenge Question

Modern deep-learning systems often process data with dimensions:

`(batch,height,width,channels)`

Explain what each dimension represents and why a matrix is insufficient for storing color images.